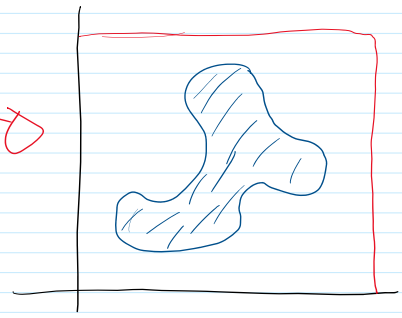
23. Intro to Volume and Integration

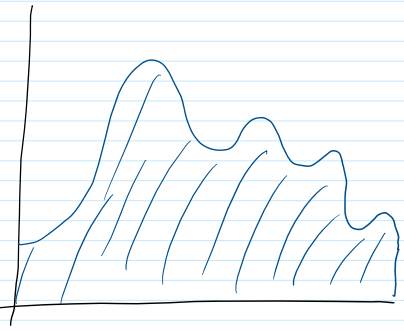
March 11, 2026 9:46 AM

What are we doing?

- Given a region in $\mathbb{R}^d$ space, we may want to estimate its volume.



- Given some function $g(x)$, $g: \mathbb{R} \to \mathbb{R}$ estimate its integral.



How are we going to do this?

- Monte Carlo Methods, sample randomly (in a smart way) to estimate these volumes and Integrals.

Why are we doing this?

- Often, when fitting a statistical Model, we are faced with a complicated Integral to solve.

→ Normalizing Constant in a distribution.

→ One source of such pain is handling posterior distributions in Bayesian Stats

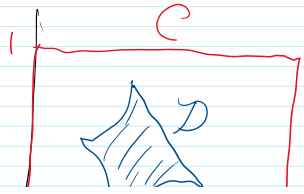
$$f(x \mid data) = \frac{f(x, data)}{f(data)}$$

Have to integrate to get such a normalizing constant

$$= \frac{f(data \mid x)\, f(x)}{f(data)}$$

Let $\mathcal{D}$ be a bounded region in $\mathbb{R}^d$

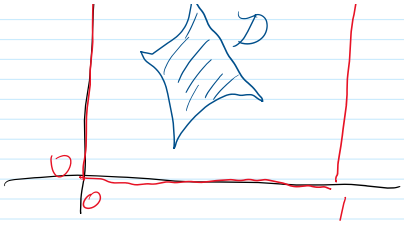
in particular, $\mathcal{D} \subset \mathcal{C} = [0,1]^{\times d}$



Define an indicator function

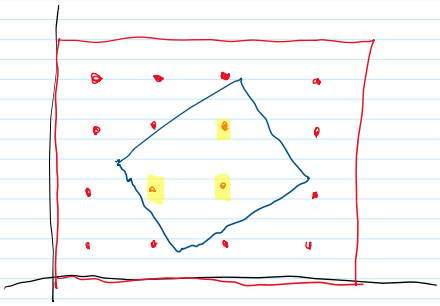
Define an indicator function

$$\phi(x) = \begin{cases} 1, & x \in \mathcal{D} \\ 0, & x \notin \mathcal{D} \end{cases}$$



$$\mathrm{Vol}(\mathcal{D}) = \int_{\mathcal{D}} dx = \int_{C} \phi(x)\, dx$$

One way to estimate $\mathrm{Vol}(\mathcal{D})$ is by creating a grid (Deterministically)



By making this grid finer i.e. adding more points, we get a better estimate of $\mathrm{Vol}(\mathcal{D})$

$$\frac{3}{16} = \frac{S}{4^2} = \frac{S}{K^d}$$

S = # points in D

K = # grid points per row

d = dimension

K^d = Total # points

Claim in the textbook is that

$$\left| \underbrace{\mathrm{Vol}(\mathcal{D})}_{\text{true volume}} - \underbrace{\frac{S}{K^d}}_{\text{estimate}} \right| \leq \frac{\overbrace{\mathrm{Surface}(\mathcal{D})}^{\text{Const}}}{\underbrace{K}_{\text{rate of convergence}}}$$

This "works" in any $\mathbb{R}^d$ space, but the # of grid points grows exponentially (Curse of Dimensionality)